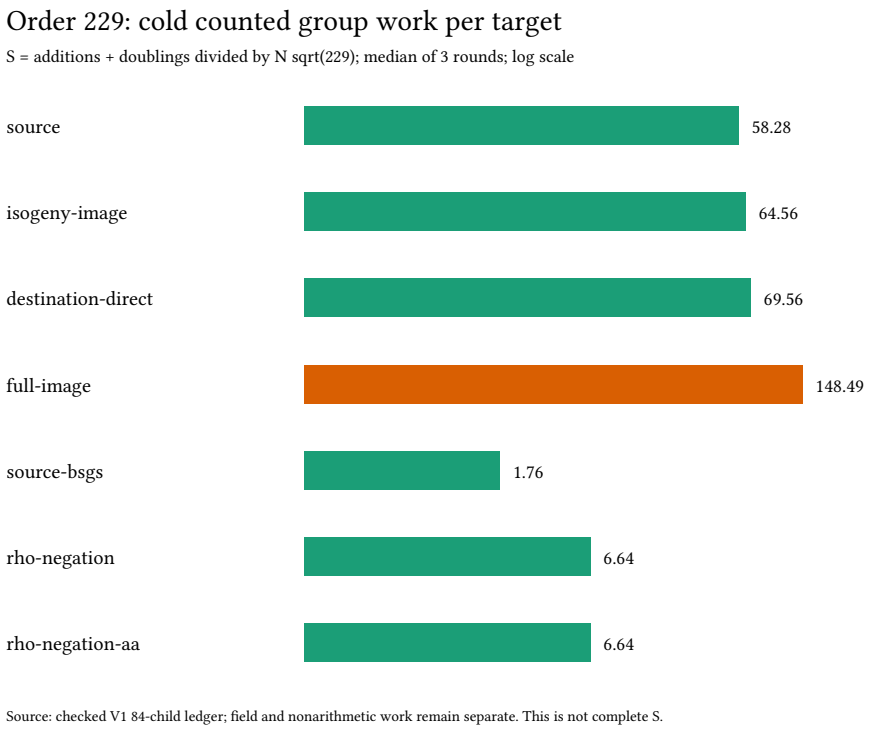
Portable cost axes and bounded order screen

V1 evidence, 2026-10-15. The native analysis re-derived 84 cold order-229 children and 1,785 independently checked scalar outputs. A separate bounded screen evaluated 30 eligible primes, accepted eight different-j rows and passed 512 independent map-witness replays. Source and binary digests, exact commands and raw records are in `FREEZE_V1.json` and `RESULTS_V1.md`.

$S_{\mathrm{group}} = \frac{\text{adds} + \text{doubles}}{N\sqrt{229}}$ uses the repository's counted group-operation convention and includes common work, setup, all queries and final scalar verification. It remains a **lower bound on complete portable cost**: the unpriced native work is named in the source report. Field multiplication and inversion counts are separate, overlapping axes.

Arm	N=1	N=4	N=16	N=64
Source projected	91.39	72.00	61.55	58.28
Isogeny-image	101.50	79.79	68.15	64.56
Destination-direct	77.98	73.83	64.84	69.56
Full rational image	5981.66	1534.12	428.12	148.49
Reusable source BSGS	4.10	2.45	1.97	1.76
Source negation rho	8.06	6.74	6.54	6.64
Rho A/A	8.06	6.74	6.54	6.64

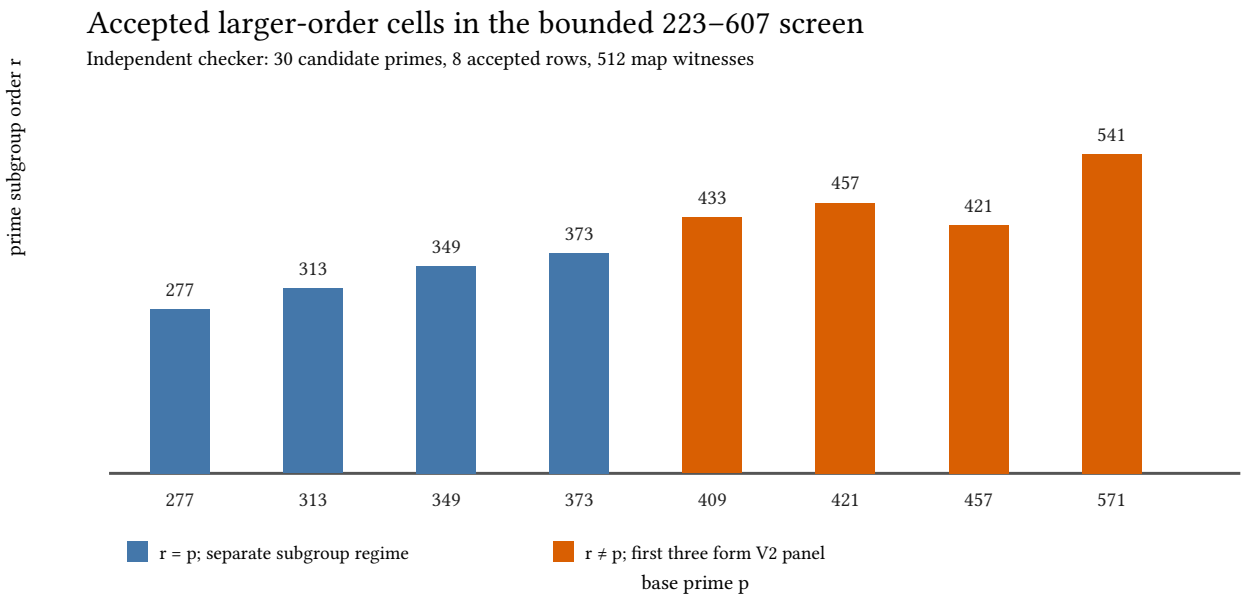
Values are median S_{group} per target across three cold rounds. At N=64, paired full-image/rho ratios are 23.402, 21.544 and 22.355; paired full-image/BSGS ratios are 85.200, 83.618 and 84.147. The full-image arm exceeded both direct controls in each counted arithmetic axis on every matched child. The generic two-class collision floor is 0.8862 per target; the present lower bounds are not full method costs.



Larger-order candidates and next gate

The V1 selection picked $r = p$ at p=277, 313 and 349. Those rows passed their arithmetic checks, but have a different subgroup regime from the order-229 reference. The versioned V2 protocol explicitly selects the first three distinct prime-to-characteristic orders from the independently checked candidate rows, before any new target generation or timing.

p	r	cofactor	j source / dest	panel state
409	433	387	84 / 368	V2 cell A
421	457	387	55 / 176	V2 cell B
457	421	495	119 / 57	V2 cell C
211	229	195	77 / 160	checked reference



The exact next step is a full rational-fiber census, independent subgroup-map replay and matched 64-target relation/recovery panel at each V2 cell. A separate L2-or-better timing session and a second-host performance replay remain required. The broader goal is an end-to-end, cold structural ECDLP method whose complete portable cost falls below same-target rho across comparable orders. Direct chi-coordinate roots and pairing transfer remain separate possible mechanisms, each requiring its own full cost and scalar replay. The frontier graph is unchanged.